

- Authentication Mode (`AuthModeEnum AuthMode`), mapping to an authentication mode, directly comparable to `AccessControlEntry AuthMode`.
- Subjects List (`list<SubjectID> Subjects`), mapping incoming message source to a type of subject, such as a CASE session Source Node ID.
- Fabric Index (`FabricIndex FabricIndex`), mapping to a fabric reference.

6.6.6.2. Overall Algorithm

The algorithm takes as input:

- the Access Control List (`acl`) from the Access Control Cluster
- the Privilege (`request_privilege`) required for the given request
- the ISD of Incoming Message (`subject_desc`)
- the Endpoint ID (`endpoint_id`) for which the querier requires a Privilege level
- the Cluster ID (`cluster_id`) for which the querier requires a Privilege level
- the Cluster Element (`cluster_element`) for which the querier requires a Privilege level
- the Request Type (`request_type`) for which the querier requires a Privilege level (ReadAttribute, ReadEvent, WriteAttribute, InvokeCommand)
- the ARL attribute (`arl_attr`) from the Access Control Cluster
- the CommissioningARL attribute (`commissioning_arl_attr`) from the Access Control Cluster

The output of the algorithm is:

- The access control status of the request, which is one of {AccessGranted, AccessRestricted, AccessDenied}.

The algorithm first determines the set of privileges granted for the Action Path, which is a subset of {View, Operate, Manage, Administer} as described in `AccessControlEntryPrivilegeEnum`. If the Privilege required for the request is contained within this set, and the request is not restricted by the Access Restriction List, then the AccessGranted status is returned. If the Privilege required for the request is contained within this set, and the request is restricted by the Access Restriction List, then the AccessRestricted status is returned. Otherwise, the AccessDenied status is returned.

The computation of the ISD is a pre-condition to the algorithm and is described in [Section 6.6.6.3, “Derivation of ISD from Incoming Message”](#).

The goal is to find the complete set of privileges granted given the input. The principle of least privilege is respected by virtue of the entire Access Control List having been computed with rules such that the least privilege is granted to all subjects. Therefore, any Access Control Entry granting the `required privilege` to the subject for a given target is sufficient to determine whether access is allowed.

The algorithm SHALL function as follows:

```
def subject_matches(acl_subject, isd_subject):
```